CSA4701 – DEEP LEARNING

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ASSIGNMENT 4 – GENERATIVE MODEL DESIGN & OUTPUT AUDIT

Assessment topic: Design and train a DBN or RBM-based model and a directed generative network for structured or unstructured data. Evaluate generated outputs for quality, diversity, structural correctness, and utility using a train-on-synthetic, test-on-real approach.

TASK 1 – Define the Aim and Data Type

Aim: To train one RBM- or DBN-based model and one directed generative model, generate synthetic examples, and audit whether the generated data are realistic, varied, structurally valid and useful for a downstream task.

Choose one data type and keep it consistent. For structured/tabular data, specify columns and valid value ranges. For images or text, define the input shape or sequence format and appropriate quality checks.

TASK 2 – Design the RBM or DBN Model

An RBM is an energy-based model with visible units representing the data and hidden units learning features. A Deep Belief Network (DBN) stacks multiple layers of learned representations. Train the selected model using the method supported by the chosen framework, and report the number of layers, units, activation functions and training settings.

If the data are binary, sigmoid units may be appropriate for visible or hidden units. For continuous data, select a model configuration suitable for the feature distribution rather than assuming binary units are suitable.

TASK 3 – Design the Directed Generative Model

A Variational Autoencoder (VAE) is one possible directed generative model. Its encoder maps data to a latent distribution and its decoder generates data from sampled latent variables. The training objective commonly combines reconstruction loss with a KL-divergence penalty: L = reconstruction loss + β ×

KL(q(z|x) || p(z)). A GAN is another possible choice, but it uses a generator and discriminator and requires a different training and evaluation description. State clearly which directed model is used.

TASK 4 – Train Both Models Fairly

1. Prepare and split the real dataset into training, validation and test sets.
2. Train each generative model using the training data only.
3. Use validation data to choose training settings and select checkpoints.
4. Generate a fixed number of synthetic examples from each trained model.
5. Keep the real test set untouched until the final evaluation.
6. Record model settings, training time, random seeds where possible, and software versions.

TASK 5 – Audit Output Quality and Diversity

Quality checks whether generated samples resemble valid examples from the real domain. Diversity checks whether generated samples differ from one another rather than repeating a few outputs. Structural correctness checks whether outputs satisfy domain rules.

- For tabular data: check missing values, allowed ranges, valid categories, impossible combinations, feature distributions and correlations.
- For images: inspect a sample grid and use a suitable distribution or feature-based metric if available.
- For text: check readability, duplicates, length and task-specific validity.
- Diversity: report duplicate rate and feature-wise or embedding-space spread where appropriate.
- Privacy/memorization: inspect whether generated examples reproduce training records too closely when this is relevant.

TASK 6 – Test Utility with Train-on-Synthetic, Test-on-Real

Train a downstream predictor using the synthetic training set and evaluate it on the untouched real test set. Compare it with a baseline predictor trained on real training data and evaluated on the same real test set. This tests whether synthetic data preserve useful patterns for the target task.

Model training data	Evaluation data	Metric	Observed result
Real training data	Real test data	Choose accuracy/F1 or MAE/RMSE	Enter measured value
Synthetic data from RBM/DBN	Real test data	Same metric as baseline	Enter measured value

Synthetic data from directed model	Real test data	Same metric as baseline	Enter measured value

Use the same downstream model, preprocessing, target definition and real test set for all comparisons. Do not treat a high synthetic-sample quality score as proof that synthetic data are useful for prediction.

TASK 7 – Record Experimental Results

Audit dimension	RBM/DBN result	Directed model result	How it was measured
Sample quality	Enter observation/metric	Enter observation/metric	Visual review or domain metric
Diversity	Enter observation/metric	Enter observation/metric	Unique rate/spread/coverage
Structural correctness	Enter observation/metric	Enter observation/metric	Domain-rule checks
Downstream utility	Enter measured metric	Enter measured metric	Train on synthetic, test on real
Training time	Enter measured value	Enter measured value	Wall-clock time
Privacy/memorization check	Enter finding if assessed	Enter finding if assessed	Duplicate/similarity audit

TASK 8 – Analyse Limitations and Defend the Final Model

Compare both models across all audit dimensions. A model may create convincing examples but lack diversity or violate domain rules. Another may generate less realistic-looking samples but preserve information useful for a downstream task. Discuss dataset size, training stability, class imbalance, possible bias, privacy concerns and the limits of each evaluation metric. Select a model only after considering the measured evidence.

TASK 9 – Conclusion and References

Generative models can create synthetic data, but generated outputs must be checked for quality, diversity, validity and usefulness. The train-on-synthetic, test-on-real approach evaluates whether the

synthetic data retain patterns needed for a practical task. Cite the dataset, framework and original or authoritative references for the selected RBM/DBN and VAE/GAN methods.